Area

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 105 minutes · Total marks: 105

KEY RULES

Rectangle	$A = l \times w$
Triangle	$A = \frac{1}{2}bh$; $\frac{1}{2}ab\sin C$
Parallelogram	$A = bh$
Trapezium	$A = \frac{1}{2}(a+b)h$
Circle	$A = \pi r^2$; sector $\frac{\theta}{360}\pi r^2$

WORKED EXAMPLE 1

[2 marks]

A triangle has base 8 cm and perpendicular height 5 cm. Work out its area.

Step 1 Area = $\frac{1}{2}\times$ base $\times$ height — $\frac{1}{2} \times 8 \times 5$

Answer: 20 cm^2

WORKED EXAMPLE 2

[3 marks]

A trapezium has parallel sides 6 cm and 10 cm and height 4 cm. Work out its area.

Step 1 Area = $\frac{1}{2}(a+b)h$ — $\frac{1}{2}(6+10) \times 4$

Step 2 Work out — $8 \times 4 = 32$

Answer: 32 cm^2

WORKED EXAMPLE 3

[3 marks]

A shape is a 10 cm by 6 cm rectangle with a semicircle of diameter 6 cm on one short side. Work out its are

Step 1 Area of rectangle — $10 \times 6 = 60$

Step 2 Area of semicircle, $r = 3$ — $\frac{1}{2} \times \pi \times 3^2 = 14.137\ldots$

Step 3 Add — $60 + 14.137 = 74.137$

Answer: 74.1 cm^2

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Area

1 A circle has a **diameter** of 14 cm.

Calculate the area of the circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 2 marks)

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2 A circle has radius 6.4 cm
Work out the area of the circle.
Give your answer correct to 3 significant figures.

..... cm^2

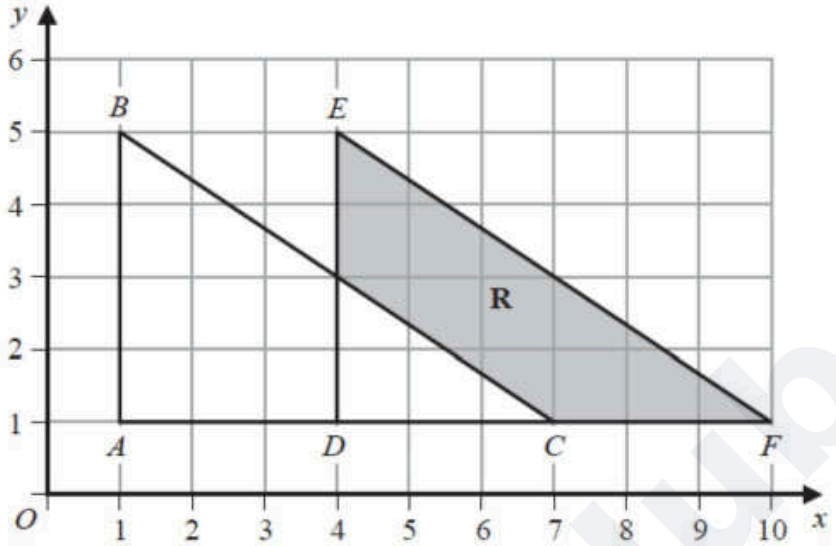
(Total for question = 2 marks)

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Area

3 The diagram shows two congruent triangles, *ABC* and *DEF*, drawn on a centimetre grid.



Find the area of the region **R**, shown shaded in the diagram.

..... cm^2

(Total for question = 3 marks)

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4 **15**

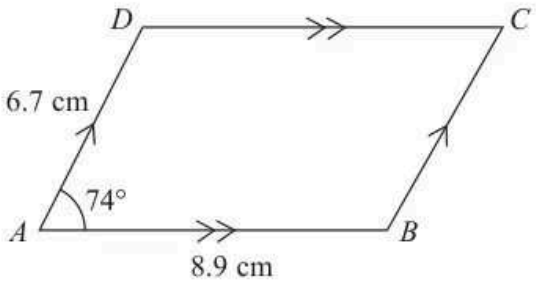


Diagram **NOT** accurately drawn

ABCD is a parallelogram.
AB = 8.9 cm.
AD = 6.7 cm.
Angle *BAD* = 74°

Calculate the area of parallelogram *ABCD*.
Give your answer correct to 3 significant figures.

..... cm²

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Area

5 *ABCD* is a kite.

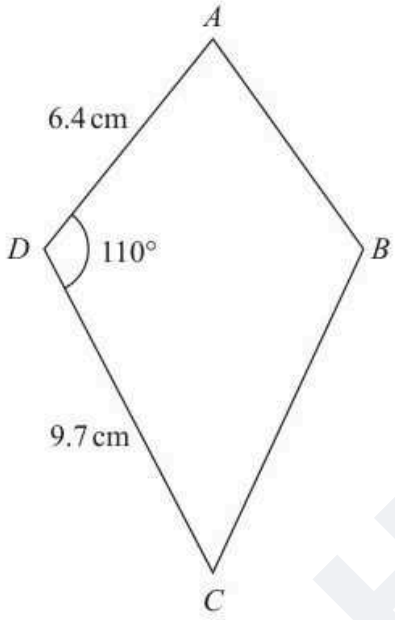


Diagram **NOT** accurately drawn

Work out the area of the kite.
Give your answer correct to 3 significant figures.

..... cm^2

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Area

6 The diagram shows a classroom wall in the shape of a trapezium.

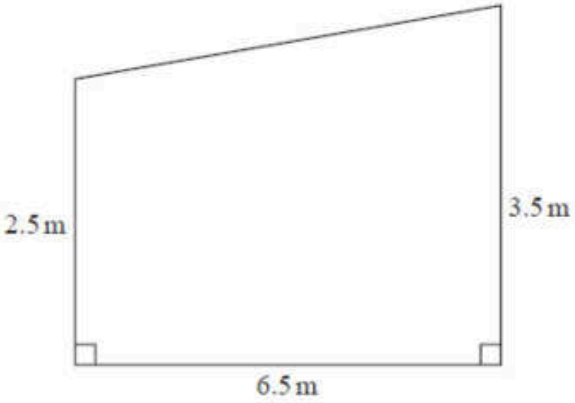


Diagram **NOT** accurately drawn

Dion wants to paint the classroom wall completely twice.
He knows that each tin of paint will cover 12 m^2

He is going to have to buy all the paint he needs.

Work out the least number of tins of paint that Dion will need to buy.
Show your working clearly.

.....

(Total for question = 4 marks)

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7 Here is a hexagon *ABCDEF*.

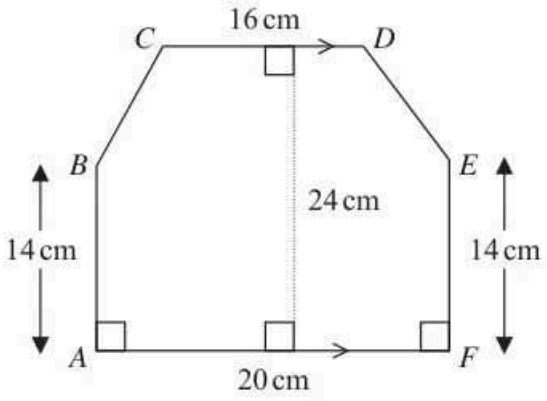


Diagram **NOT** accurately drawn

CD is parallel to *AF*.

Work out the area of hexagon *ABCDEF*.

..... cm²

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8 The diagram shows a shape.

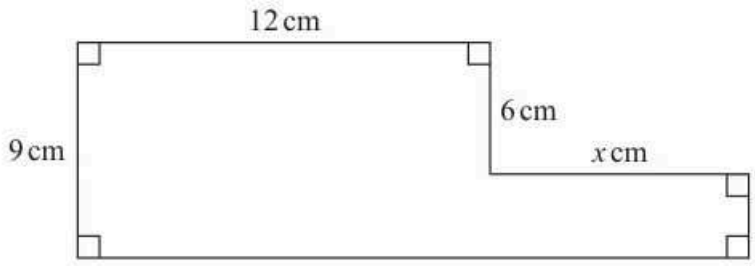


Diagram **NOT** accurately drawn

The shape has area 129 cm^2

Work out the value of x.

$x =$ .....

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9 The diagram shows a shape with one line of symmetry.

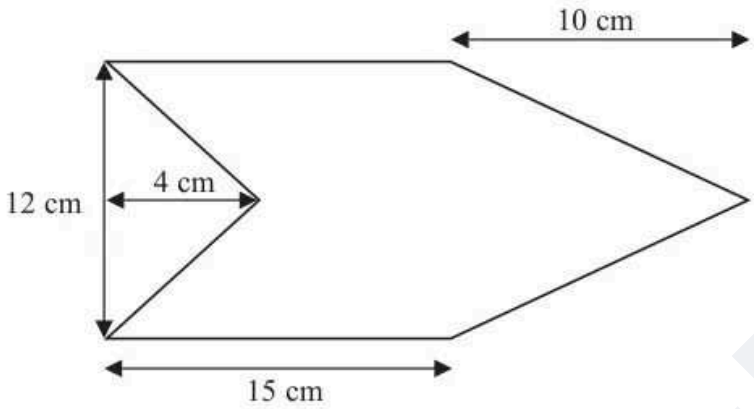


Diagram **NOT** accurately drawn

Work out the area of the shape.

..... cm^2

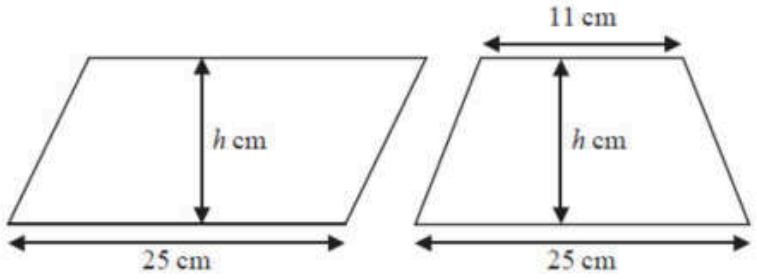
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Area

10 The diagram shows a parallelogram and a trapezium.

Diagram **NOT** accurately drawn



Both shapes have a base of 25 cm and a perpendicular height of h cm

The area of the parallelogram is 175 cm^2

Work out the area of the trapezium as a percentage of the area of the parallelogram.

.....%

(Total for question = 4 marks)

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11 The diagram shows an 8-sided shape *ABCDEFGH*.

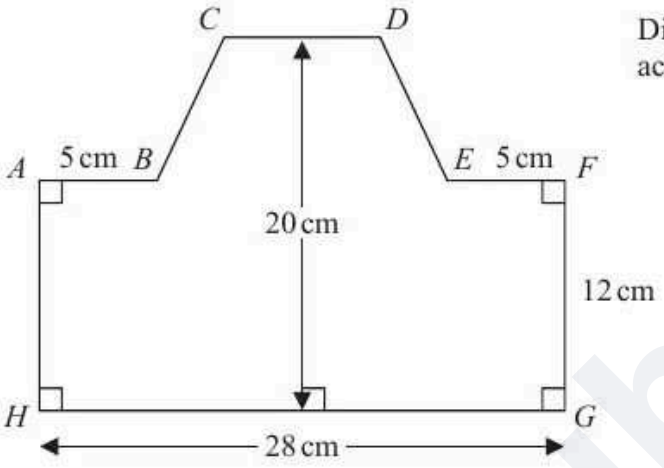


Diagram **NOT** accurately drawn

$HG = 28\,\text{cm}$ $FG = 12\,\text{cm}$ $AB = EF = 5\,\text{cm}$
The height of the shape is 20 cm
CD is parallel to *HG*

The area of shape *ABCDEFGH* is $434\,\text{cm}^2$

Find the length of *CD*.

..... cm

(Total for Question 5 is 4 marks)

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12

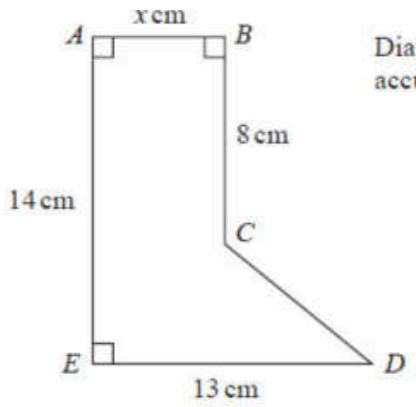


Diagram **NOT** accurately drawn

The diagram shows the shape *ABCDE*.
The area of the shape is 91.8 cm^2

Work out the value of x

x = ...

(Total for question = 4 marks)

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13 The diagram shows a 6-sided shape $ABCDEF$

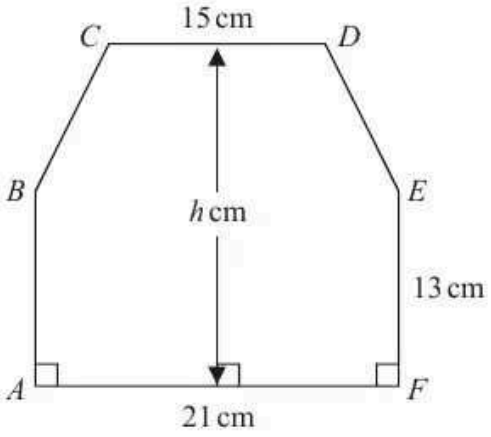


Diagram **NOT** accurately drawn

$AF = 21$ cm $\quad CD = 15$ cm $\quad AB = FE = 13$ cm

The perpendicular height of the shape is h cm
CD is parallel to AF

The area of the shape is $390\,\text{cm}^2$

Work out the value of h

$h =$ .....

(Total for Question 7 is 4 marks)

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14 7

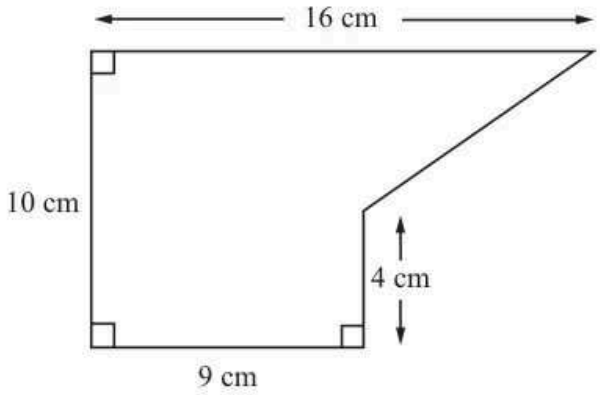


Diagram **NOT** accurately drawn

The diagram shows a shape.

Work out the area of the shape.

..... cm^2

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15 The diagram shows an isosceles triangle.

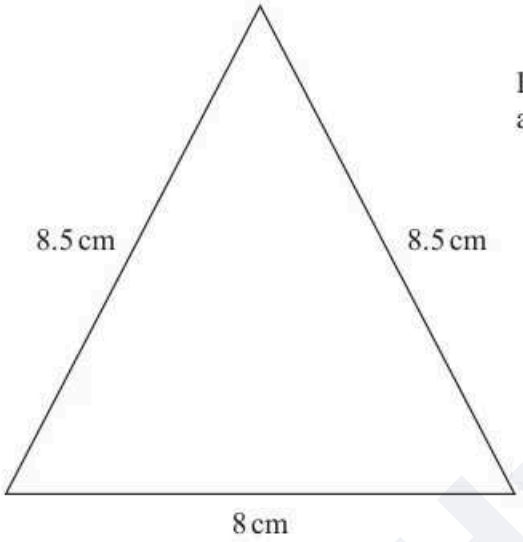


Diagram **NOT** accurately drawn

Work out the area of the triangle.

..... cm^2

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Area

16 The diagram shows isosceles triangle *ABC*

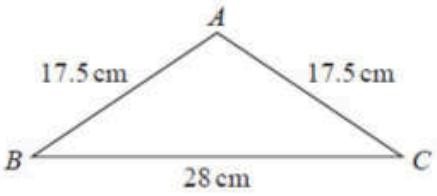


Diagram **NOT** accurately drawn

AB = *AC* = 17.5 cm *BC* = 28 cm

Calculate the area of triangle *ABC*

..... cm^2

(Total for question = 4 marks)

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Area

17 The diagram shows the front of a wooden door with a semicircular glass window.

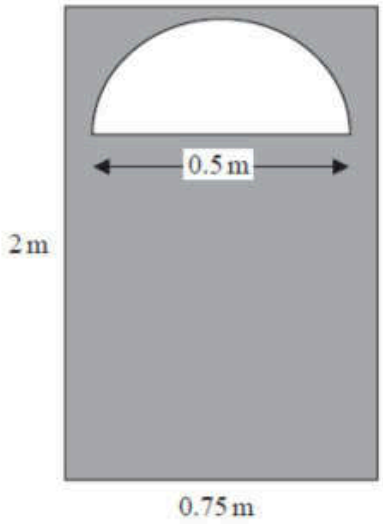


Diagram **NOT** accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

(Total for question = 5 marks)

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18 The diagram shows a shape made from a right-angled triangle and a semicircle.

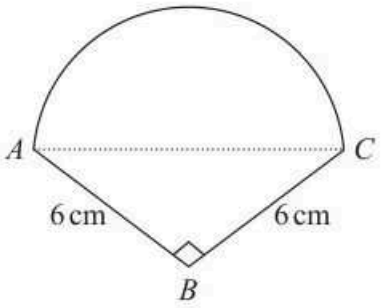


Diagram **NOT** accurately drawn

AC is the diameter of the semicircle.
$BA = BC = 6$ cm
Angle $ABC = 90°$

Work out the area of the shape.
Give your answer correct to 1 decimal place.

..... cm²

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19 **10** The diagram shows a hexagon $ABCDEF$

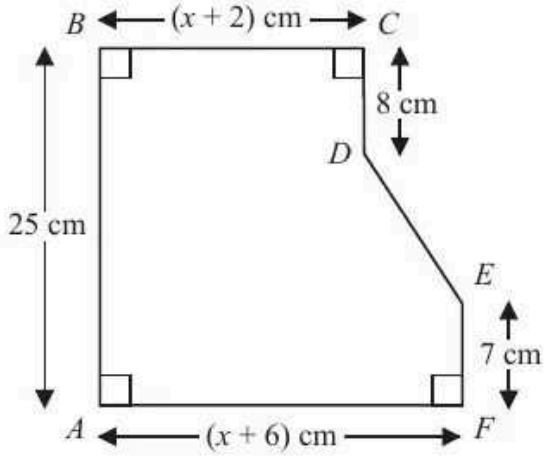


Diagram **NOT** accurately drawn

$AB = 25$ cm $\quad BC = (x + 2)$ cm $\quad CD = 8$ cm $\quad EF = 7$ cm $\quad AF = (x + 6)$ cm

The area of hexagon $ABCDEF$ is 258 cm²

Work out the value of x

$x =$ .....

(Total for Question 10 is 5 marks)

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20 **12**

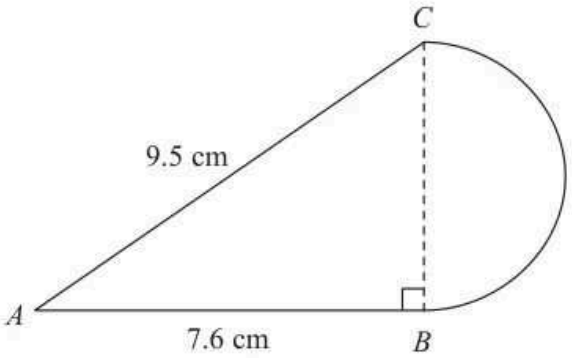


Diagram **NOT** accurately drawn

The diagram shows a shape made from triangle ABC and a semicircle with diameter BC.
Triangle ABC is right-angled at B.
$AB = 7.6$ cm and $AC = 9.5$ cm.

Calculate the area of the shape.
Give your answer correct to 3 significant figures.

..... cm^2

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CHALLENGE QUESTIONS

21 The diagram shows a pentagon.

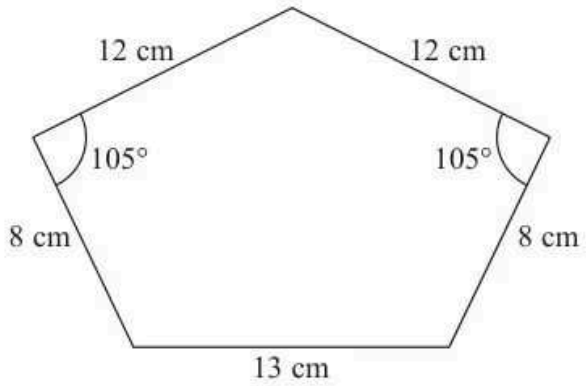


Diagram **NOT** accurately drawn

Work out the area of the pentagon.
Give your answer correct to 3 significant figures.

..... cm^2

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22

9

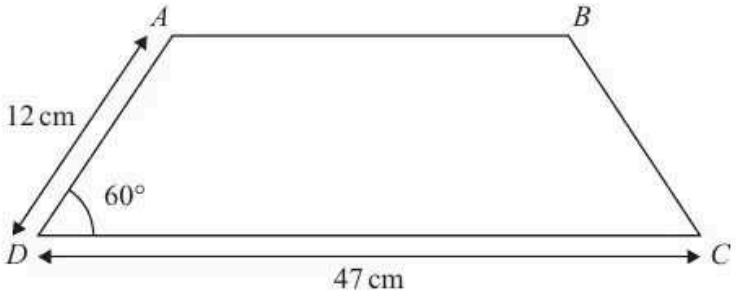


Diagram **NOT** accurately drawn

ABCD is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12\,\text{cm}$ $DC = 47\,\text{cm}$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2

(Total for Question 9 is 5 marks)

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23 **10** The diagram shows a hexagon *ABCDEF*

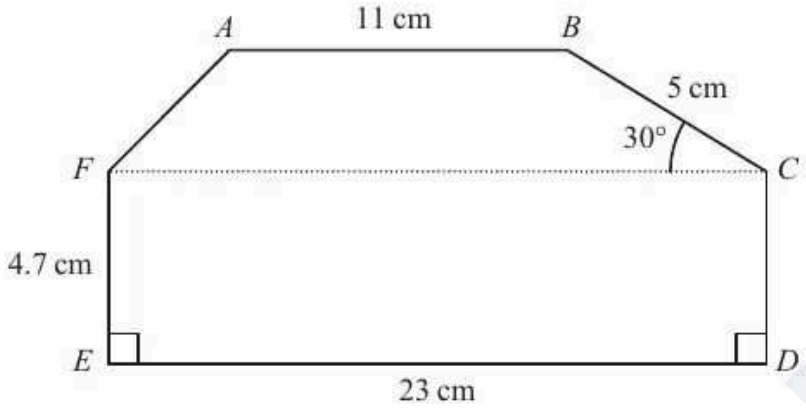


Diagram **NOT** accurately drawn

Angle $BCF = 30°$
AB, FC and ED are parallel.

Calculate the area of *ABCDEF*
Show your working clearly.

..... cm²

(Total for Question 10 is 5 marks)

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24 The diagram shows a regular octagon *ABCDEFGH*.

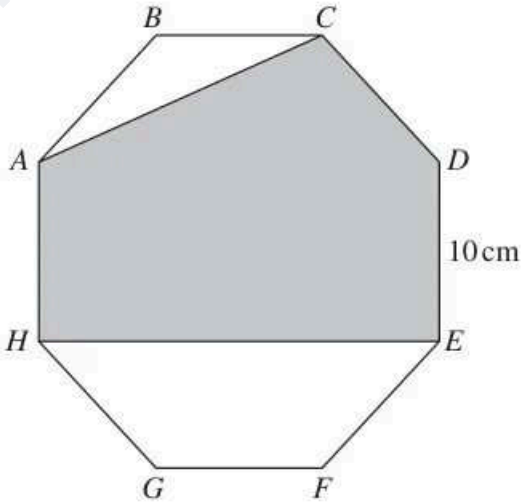


Diagram **NOT** accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region *ACDEH*.
Give your answer correct to the nearest cm²

25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

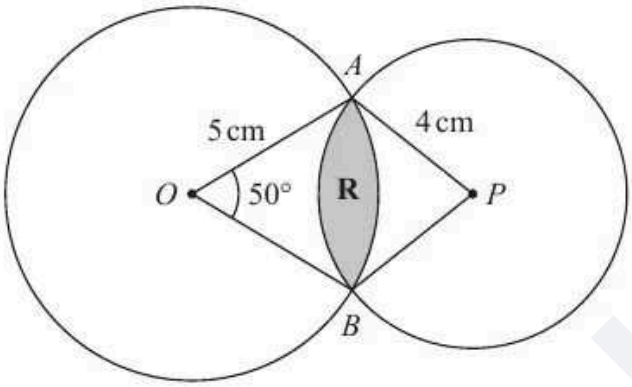


Diagram **NOT** accurately drawn

One of the circles has centre O and radius 5 cm.
The other circle has centre P and radius 4 cm.
Angle $AOB = 50°$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 25 is 6 marks)

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TOTAL FOR PAPER IS 105 MARKS

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